



ROAD HERO



Smart Road Hazard Detection &
Community Safety System

Developed By

Name	ID
Md. Asif Iqbal Emon	2311502042
Touhid Ara Himu	2312187642
Farjana Shakh Alve	2013609042



WHAT IS ROAD HERO?

Road Hero is a community-driven road safety application that empowers citizens to actively participate in making roads safer for everyone.

Using advanced AI technology, the app detects, reports, and monitors road hazards in real-time, creating a comprehensive safety network powered by the community.



THE PROBLEM

01

Poor road conditions cause thousands of accidents annually, putting drivers and pedestrians at serious risk of injury and property damage.

02

No real-time hazard reporting system exists for drivers to warn others about sudden road dangers like potholes, debris, or flooding.

03

Road damage tracking remains fragmented with no centralized platform connecting citizens, authorities, and maintenance teams.





OUR SOLUTION

01

AI-based hazard detection that automatically identifies and classifies road damage using advanced machine learning algorithms

02

Real-time hazard map tracking with live updates showing dangerous road conditions and safer alternative routes

03

Community verification system where users validate hazard reports, ensuring accurate and up-to-date safety information





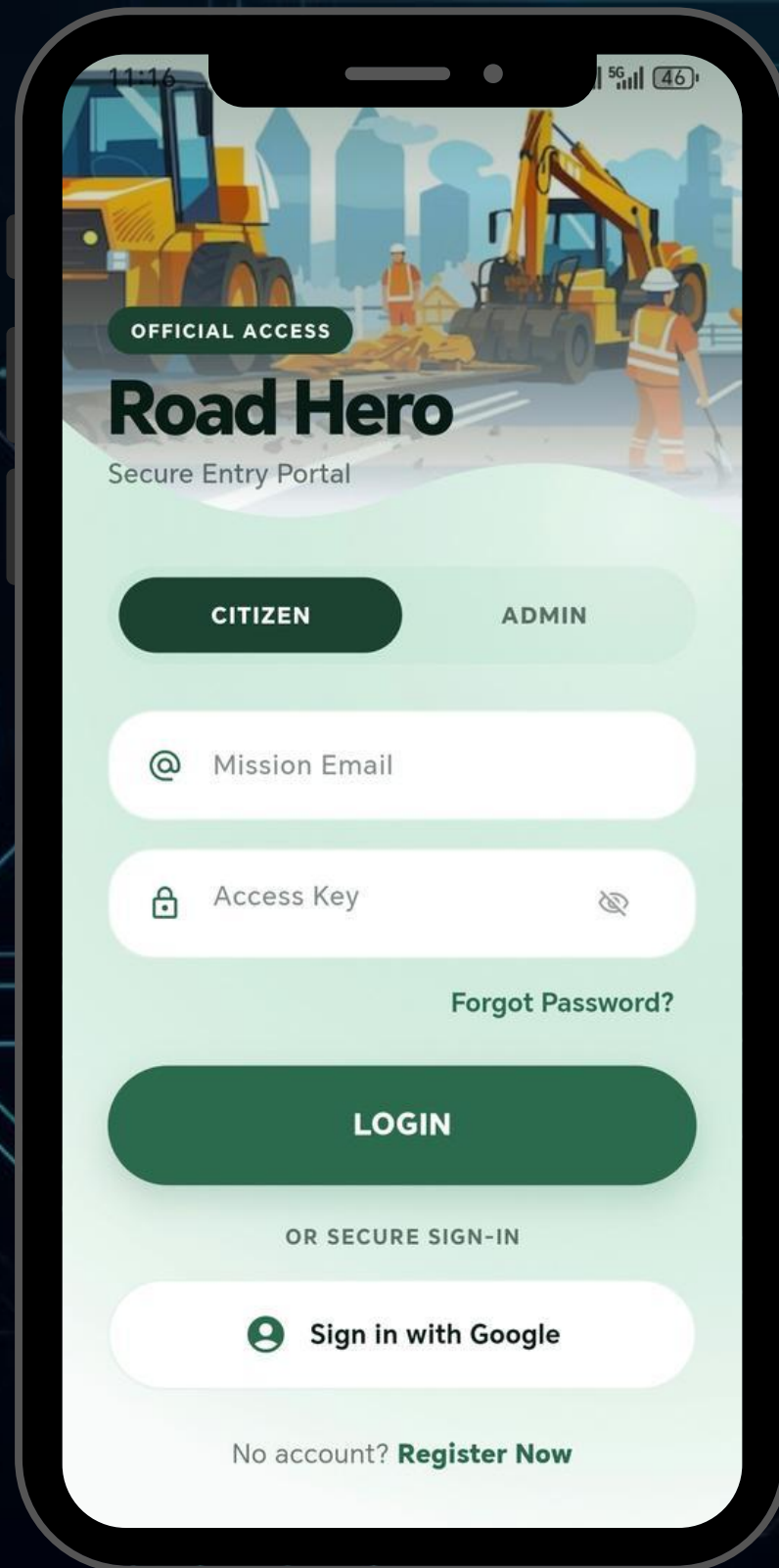
USER & ADMIN LOGIN

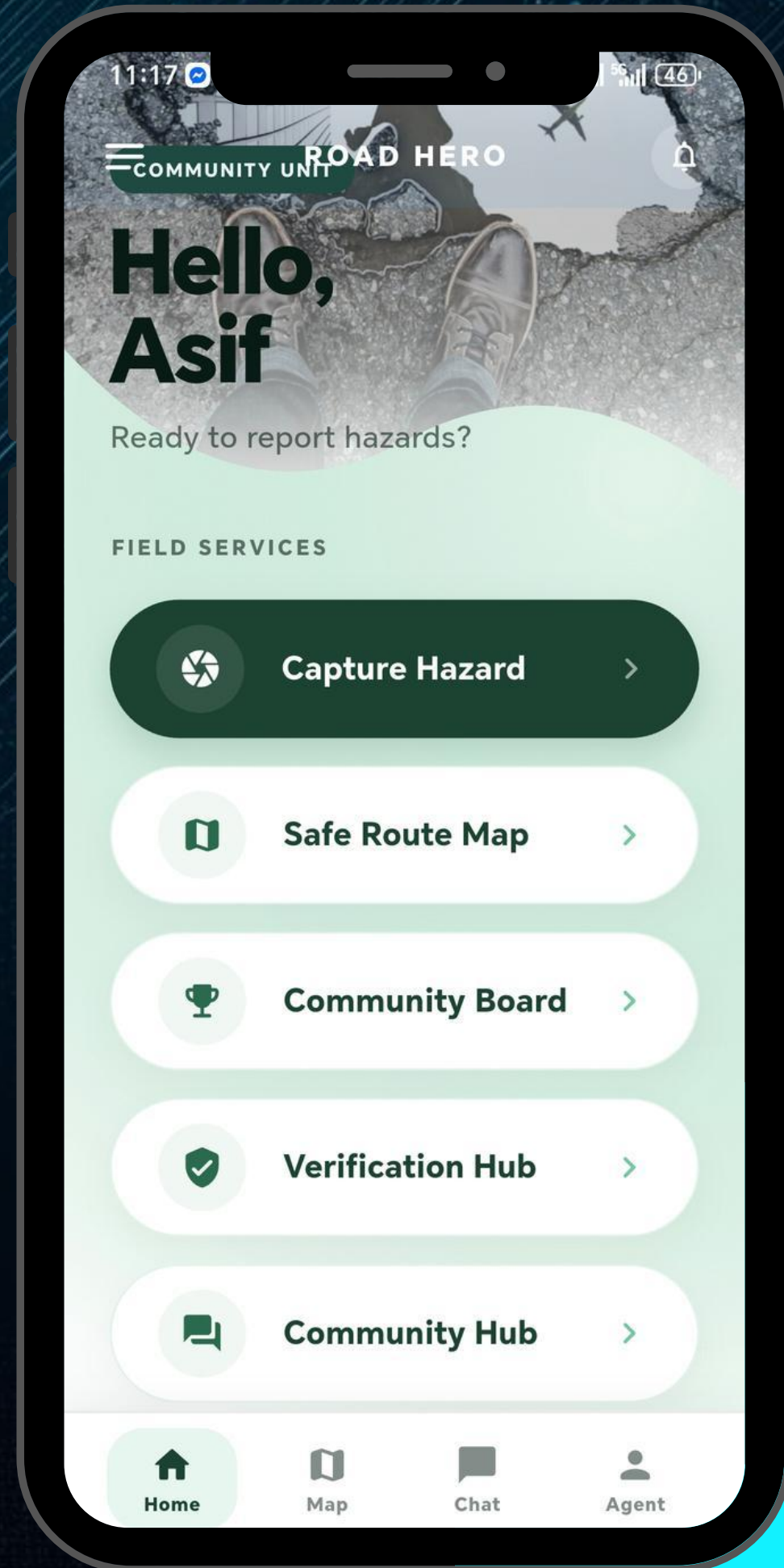
01

Separate User and Admin login portals provide tailored access levels. Users can report hazards and engage with the community, while Admins manage reports and monitor system-wide activity through dedicated dashboards.

02

Secure signup and authentication powered by modern encryption standards. Multi-factor authentication options ensure account safety while maintaining a seamless login experience for all users.





HOMEPAGE OVERVIEW

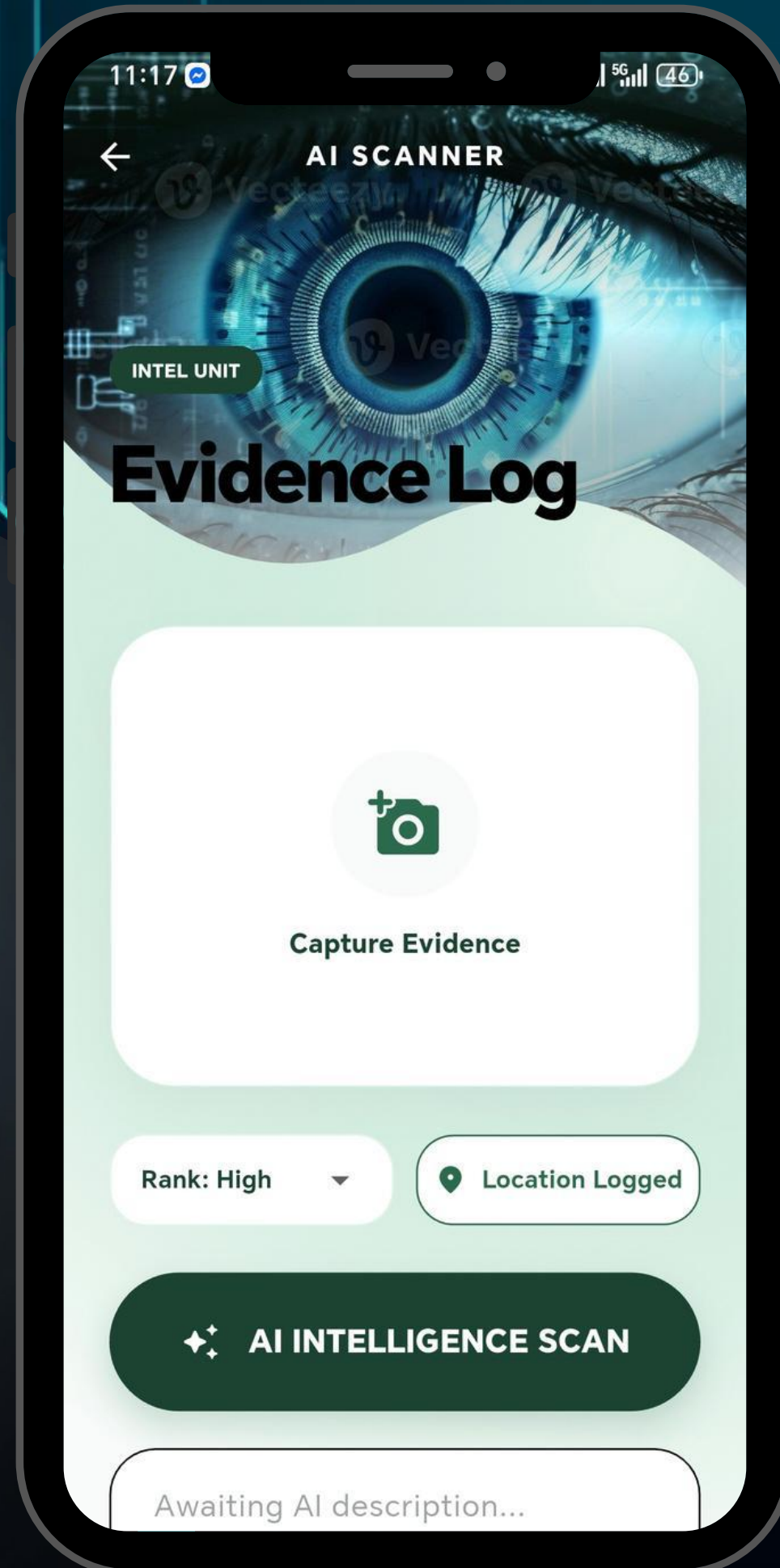
01

Access to 5 main features from your central dashboard:

- Capture Hazard - Report road damage instantly
- Safe Route Map - Navigate around hazards
- Community Board - Earn points and rankings
- Verification Hub - Validate hazard reports
- Community Hub - Engage with other users

02

The homepage serves as your command center for road safety. Quick access tiles let you jump directly to any feature. Real-time notifications keep you informed of nearby hazards. Your contribution stats and community ranking are always visible at a glance.



CAPTURE HAZARD

01

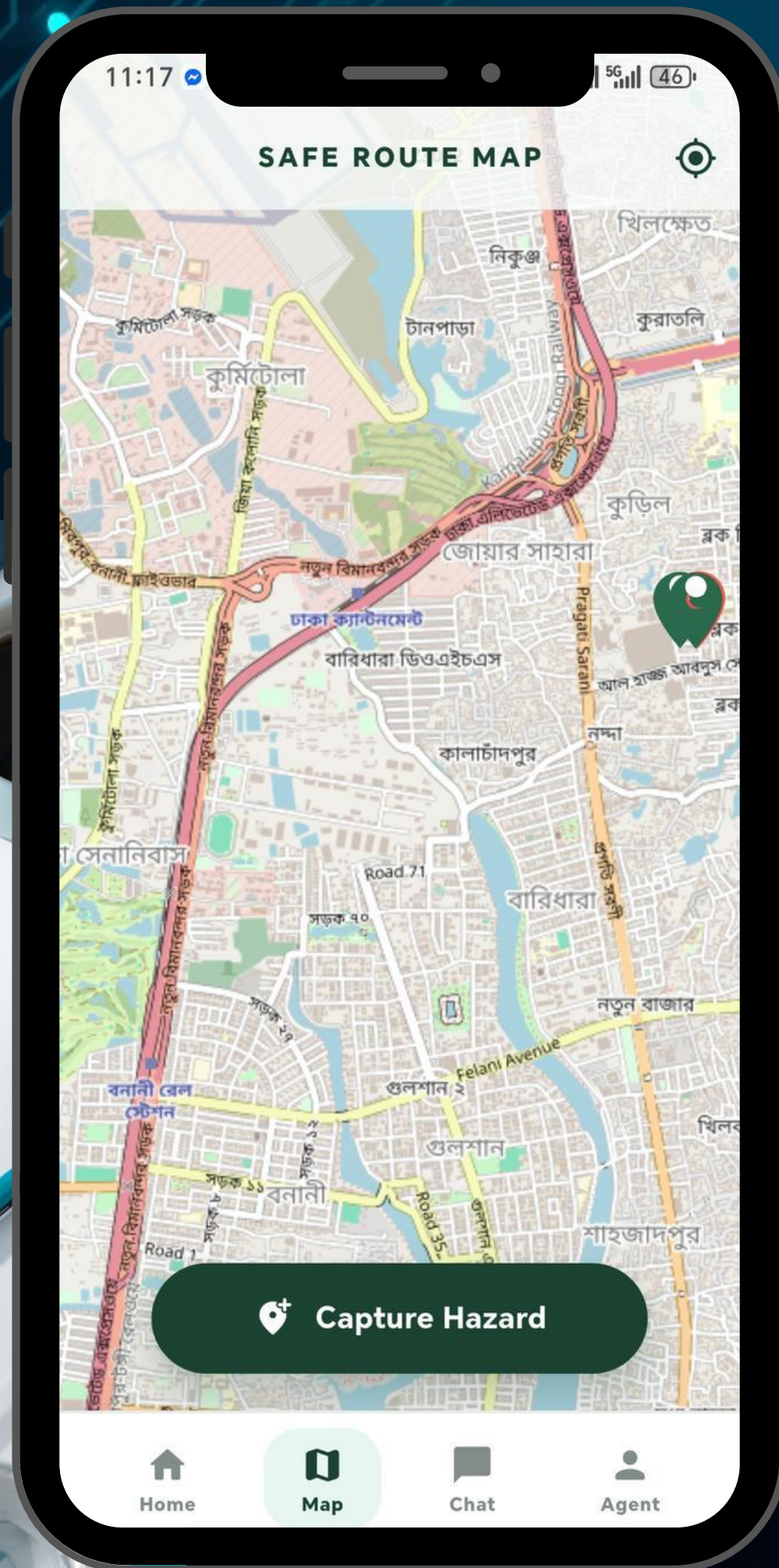
Capture road damage instantly using your phone's camera. The app automatically detects your GPS location and timestamps the report for accurate hazard mapping.

02

AI-powered analysis identifies hazard type and severity level. Reports are automatically posted to the Community Hub, Safe Route Map, and Verification Hub for community action.



Road Hero



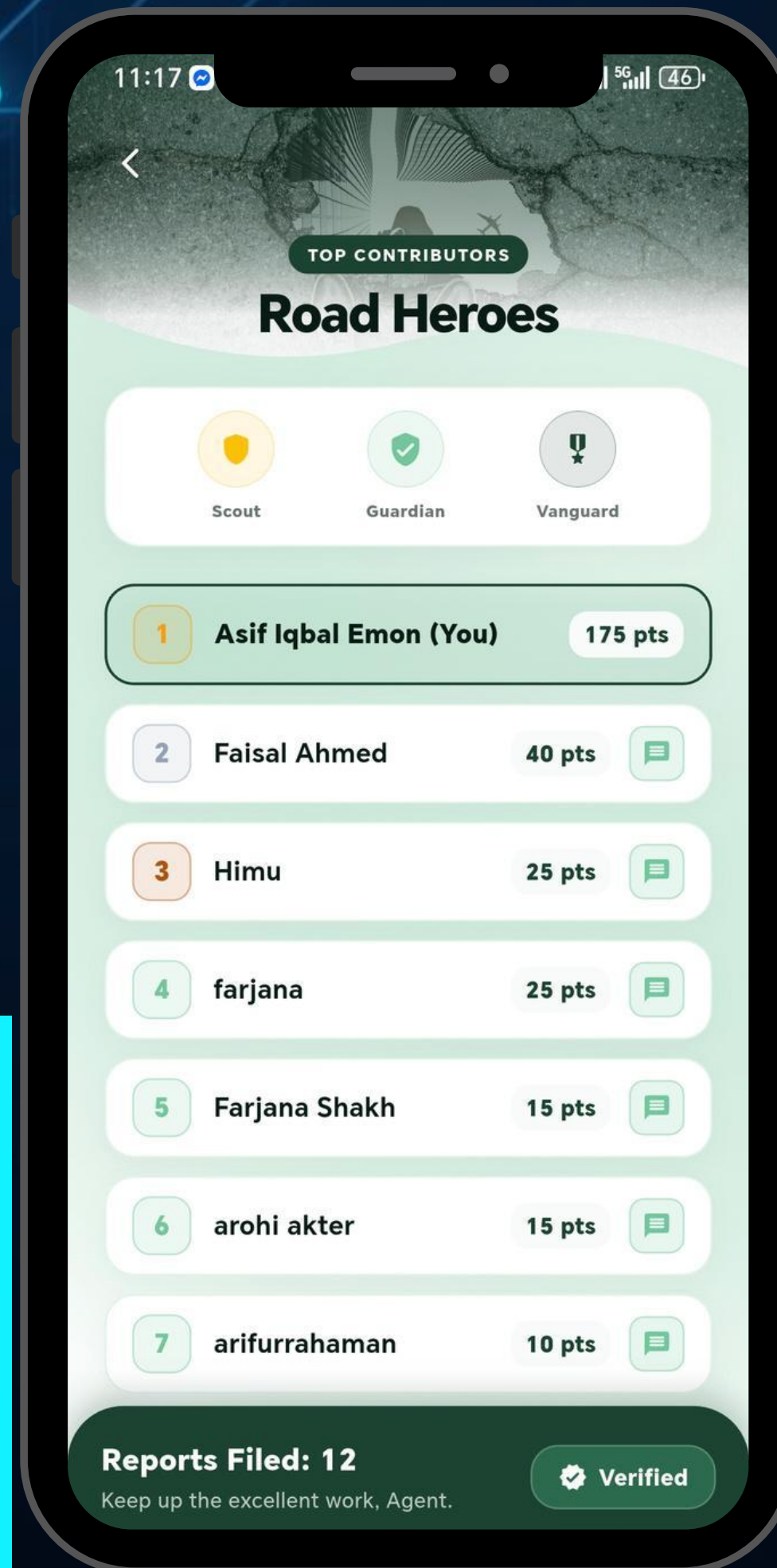
SAFE ROUTE MAP

01

- Visual hazard pins displayed on interactive map
- Color-coded markers indicate hazard severity
- Helps users select safer travel routes
- Real-time updates as hazards are reported

02

- Navigate around dangerous road conditions
- Plan trips with hazard awareness
- Avoid accident-prone areas
- Community-verified hazard locations



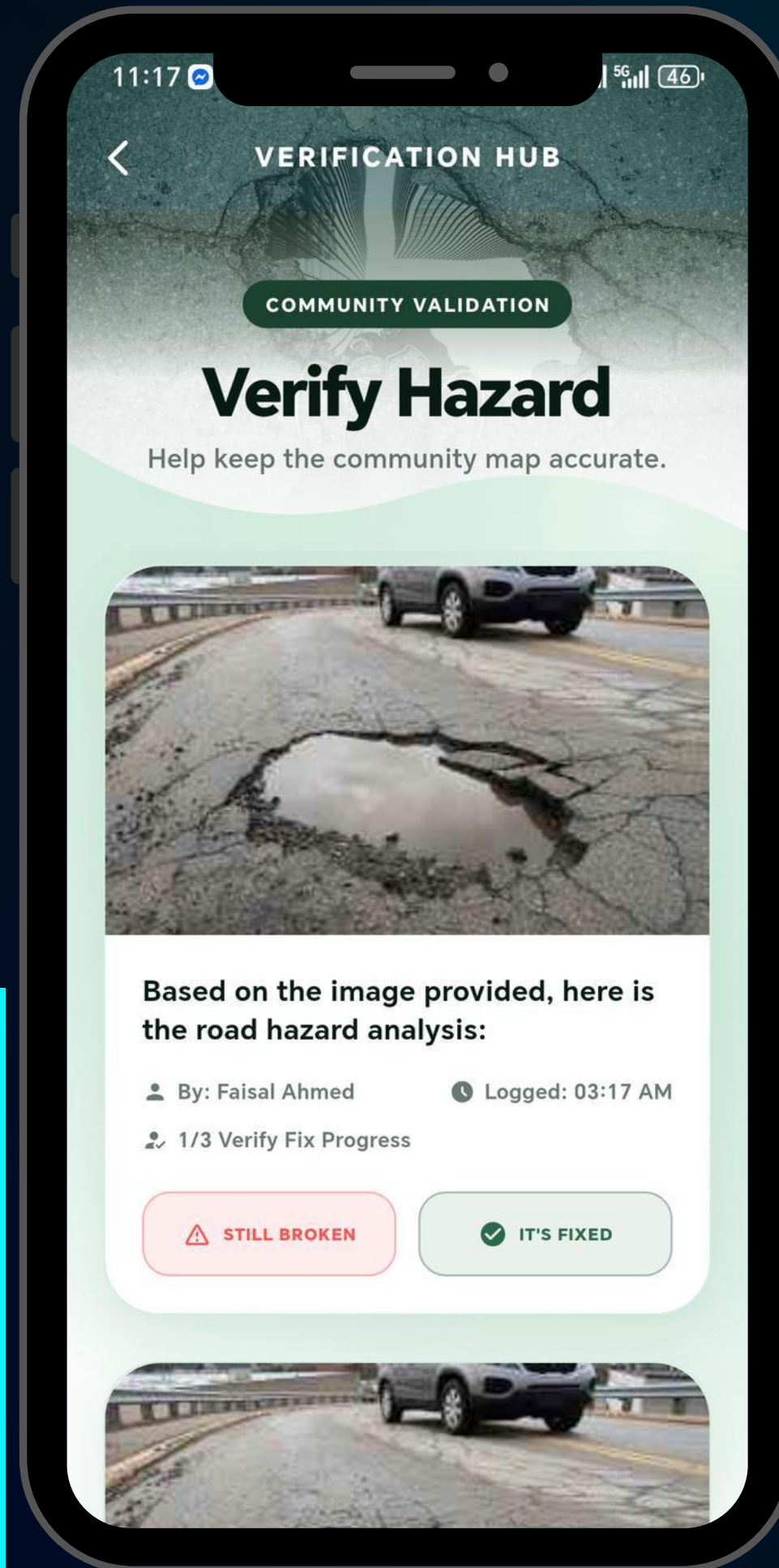
COMMUNITY BOARD

01

Earn points for every hazard you report, verify, or help resolve. Your contributions make roads safer and build your reputation in the community.

02

Compete with fellow Road Heroes on the leaderboard. Top contributors are recognized and rewarded, creating motivation for active community participation.



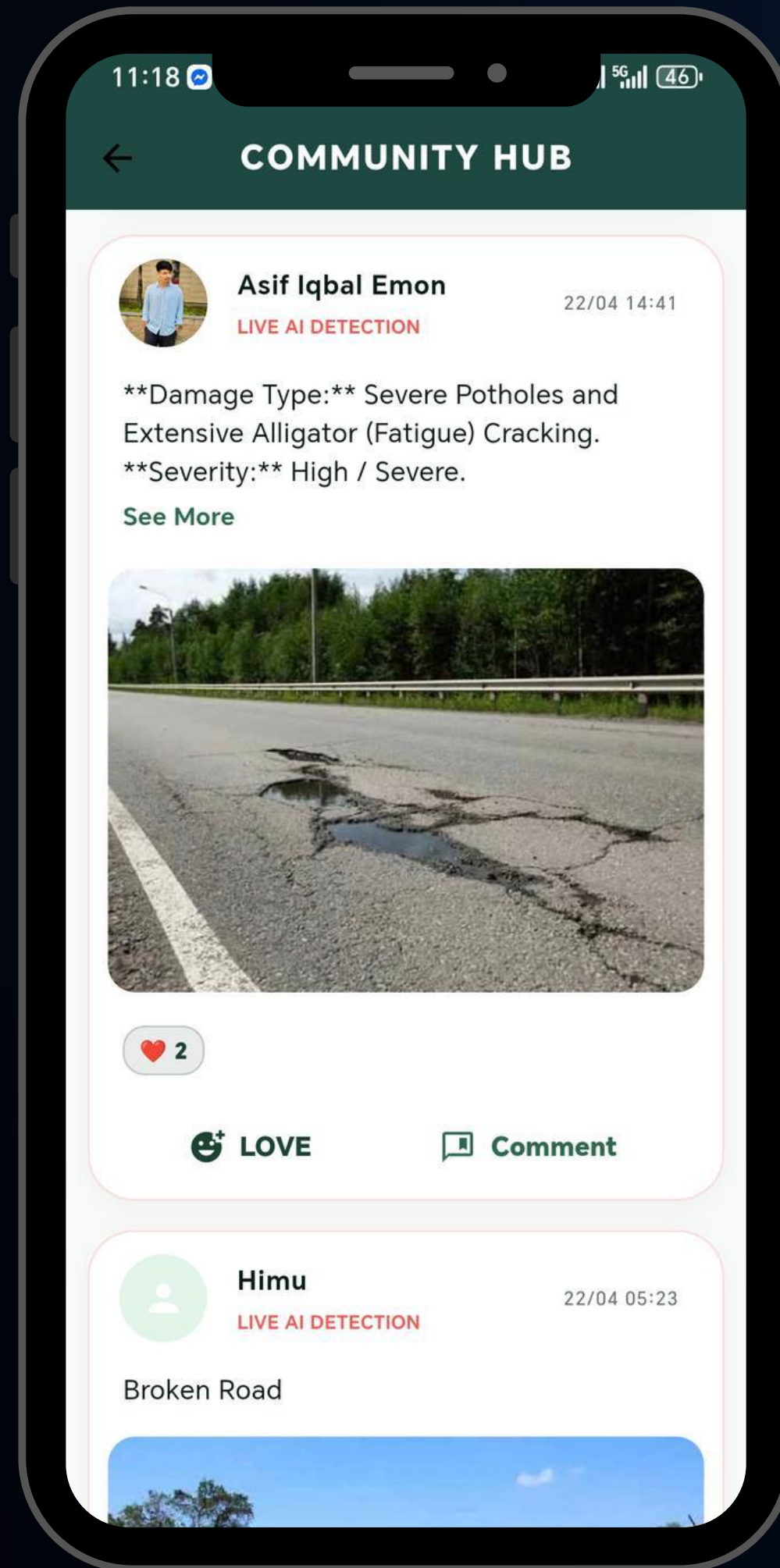
VERIFICATION HUB

01

Mark hazards as Resolved or Still Broken based on current road conditions. Community members validate reports to ensure accuracy and maintain an up-to-date hazard database.

02

Hazards are automatically removed from the system after receiving 3 resolved votes from verified users. This crowdsourced validation ensures only active hazards remain visible on the map.



COMMUNITY HUB

01

Share and discover road hazards through an interactive social feed. Post hazard reports with photos, react to community submissions, and engage through comments to keep your neighborhood informed and safe.

02

Build community awareness by interacting with hazard posts. Like, comment, and share reports to increase visibility and help prioritize dangerous road conditions in your area.



CHAT SYSTEM

01

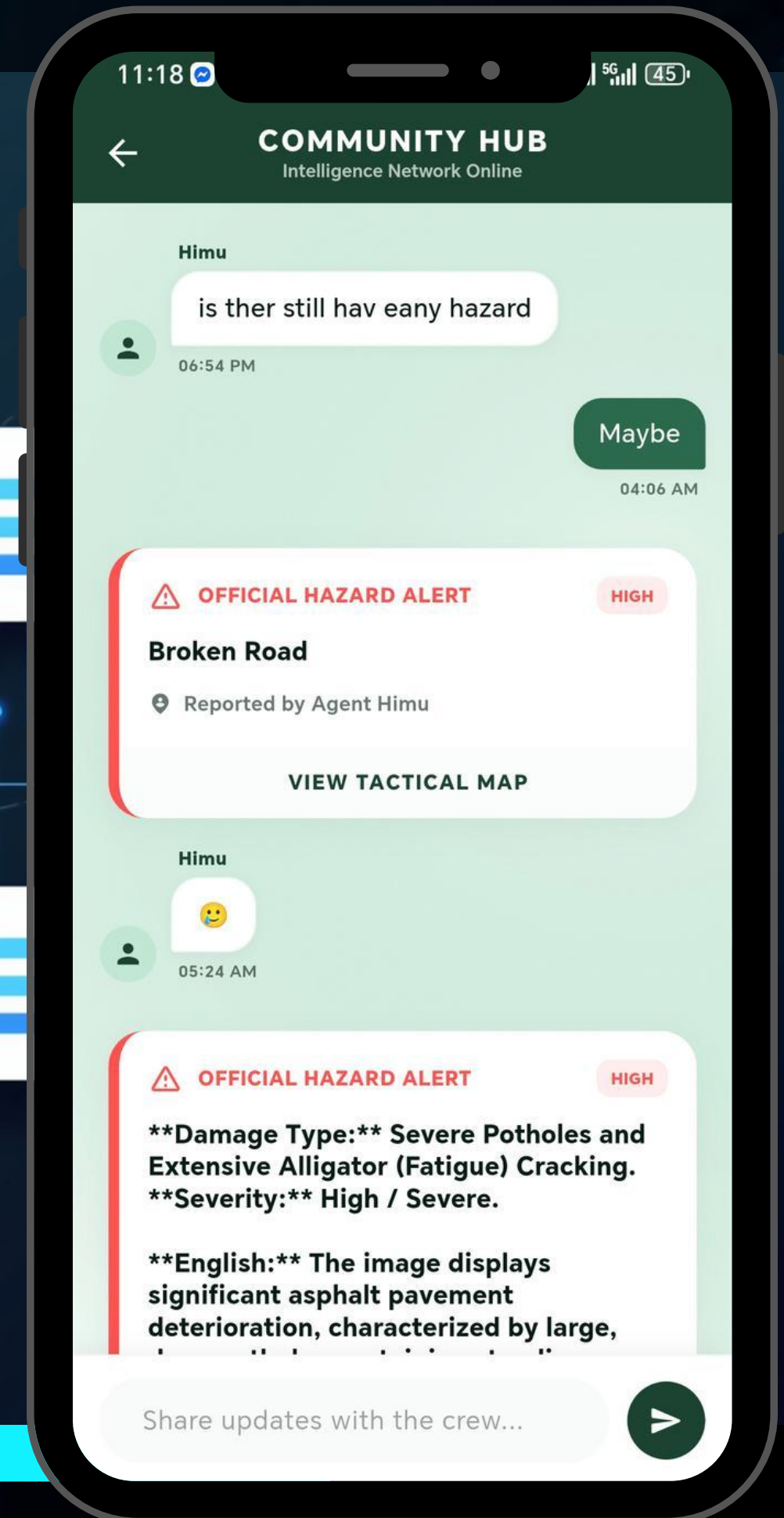
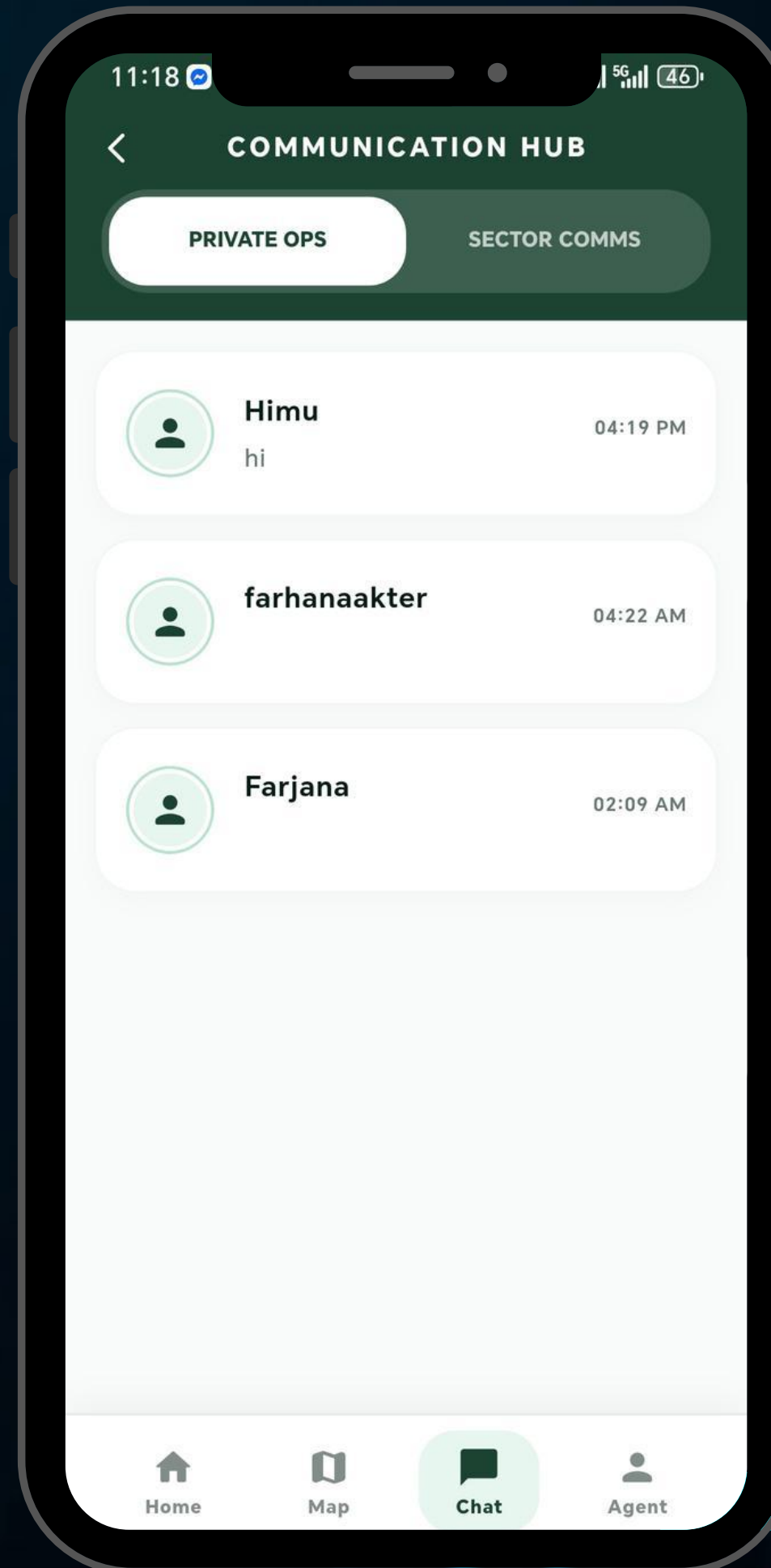
One-to-one private messaging between users for direct communication about hazards and safety concerns

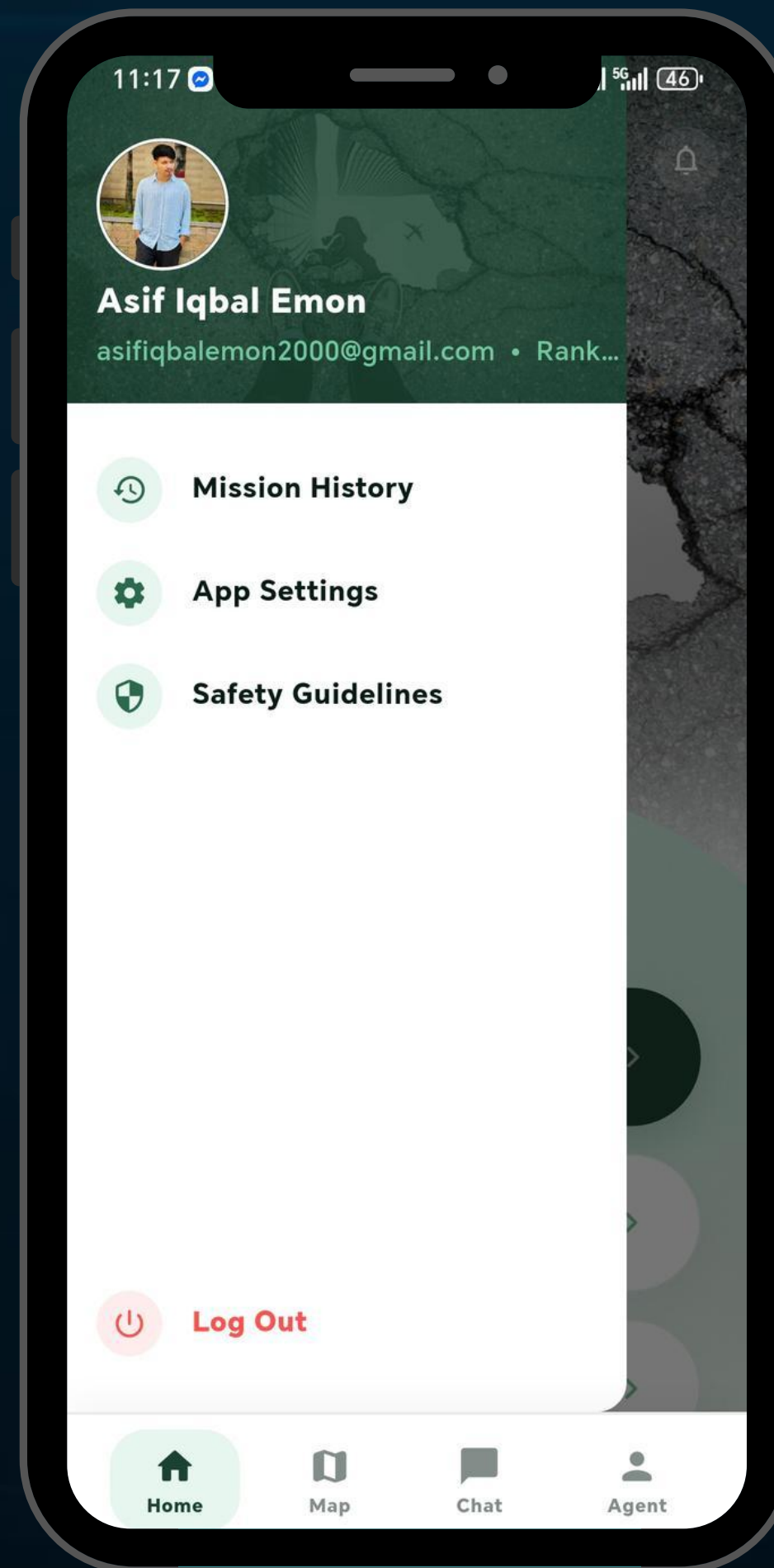
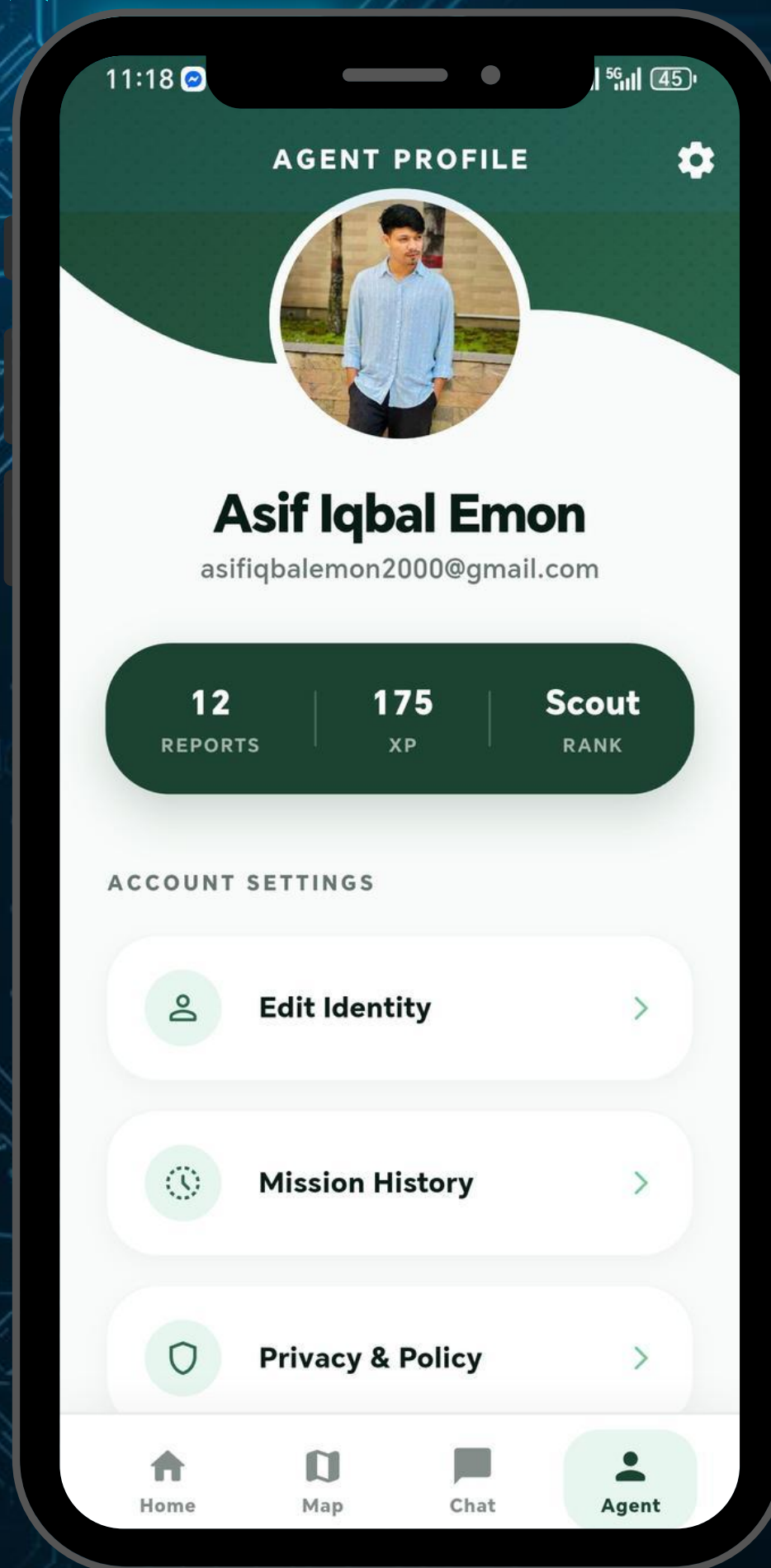
02

Community group chat for neighborhood discussions and real-time hazard alerts

03

Instant notifications and seamless communication to coordinate safety responses





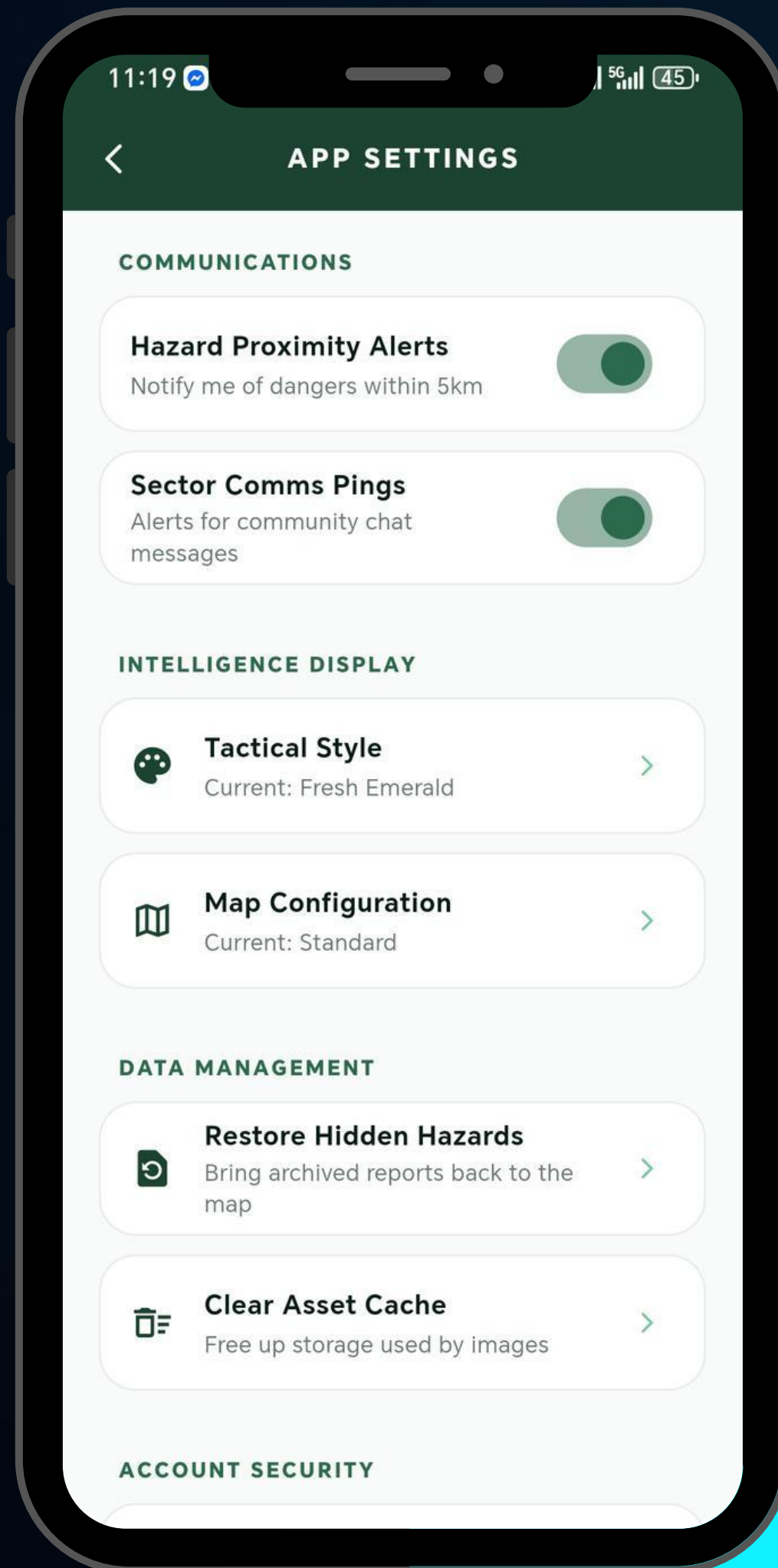
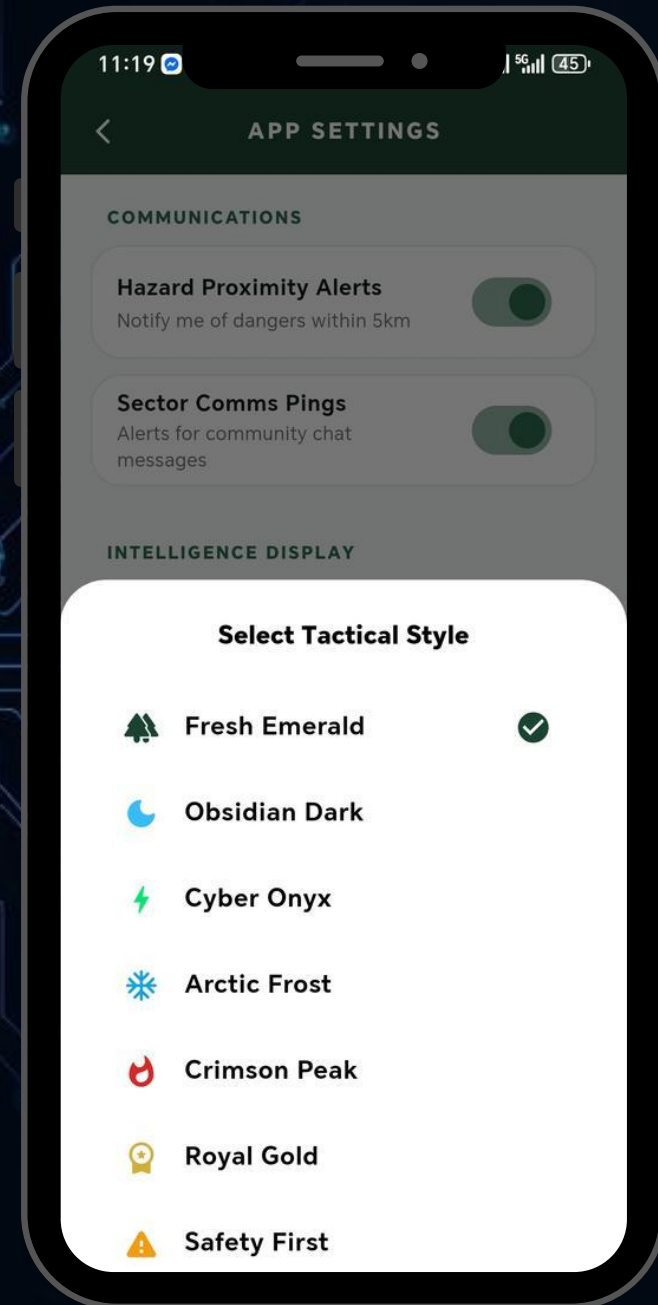
AGENT PROFILE

01

- User photo & email display
- Edit Identity options
- Mission History tracking
- Privacy & Policy access

02

Personal dashboard for managing user information, viewing contribution history, and accessing account settings.



SETTINGS MENU

01

Theme Customization

Personalize your Road Hero experience with light and dark mode options. Choose accent colors and adjust visual preferences to match your style while maintaining optimal readability for safe navigation.

02

App Options

Configure notification preferences, privacy settings, and data management. Control location permissions, manage storage, and customize how Road Hero interacts with your device for the best user experience.



Road Hero



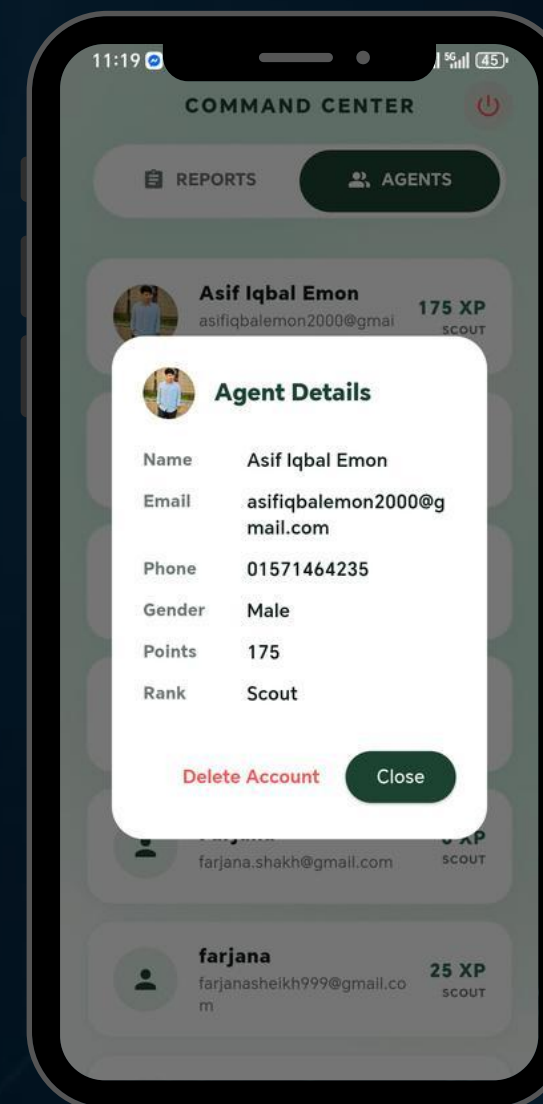
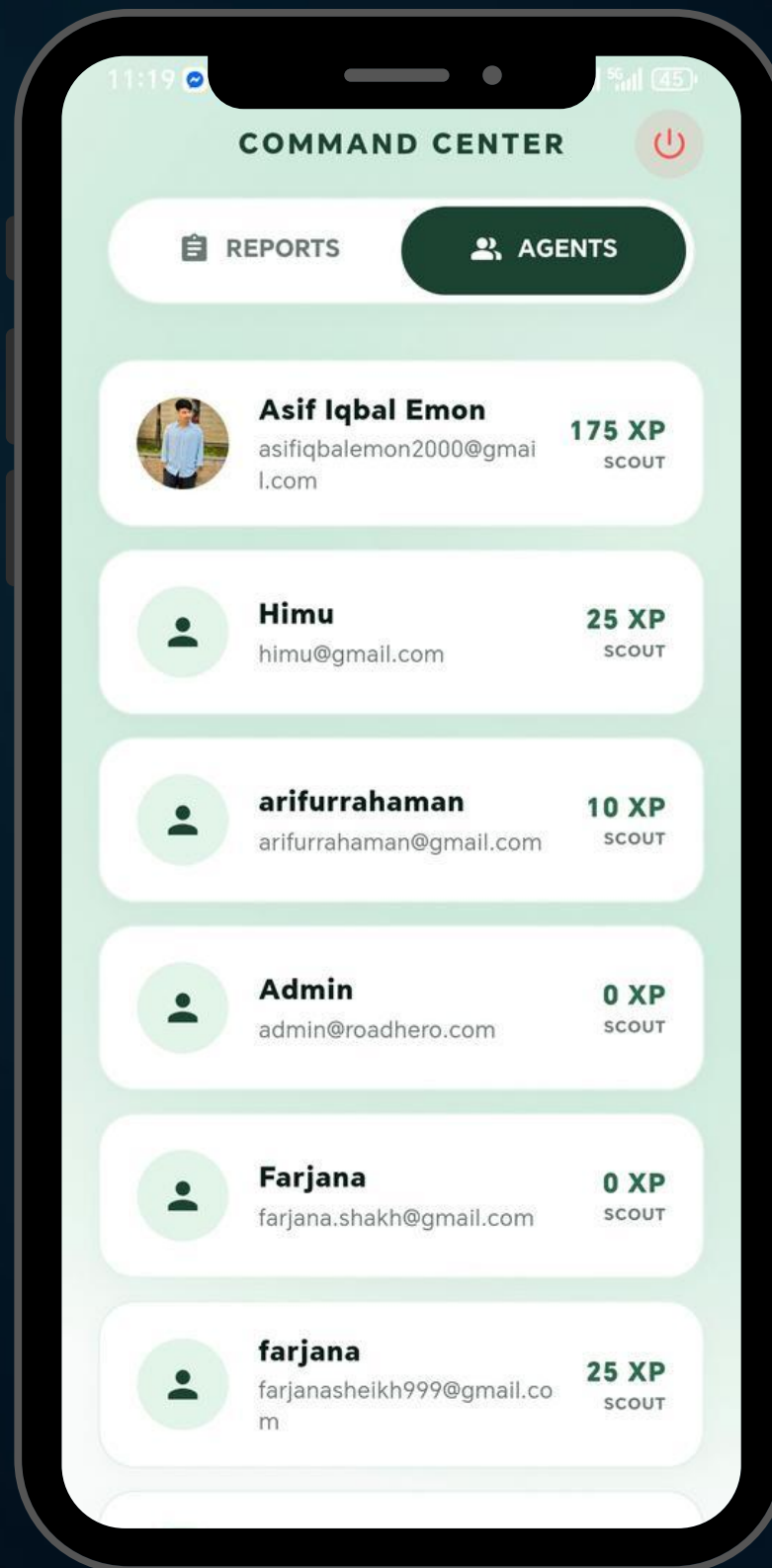
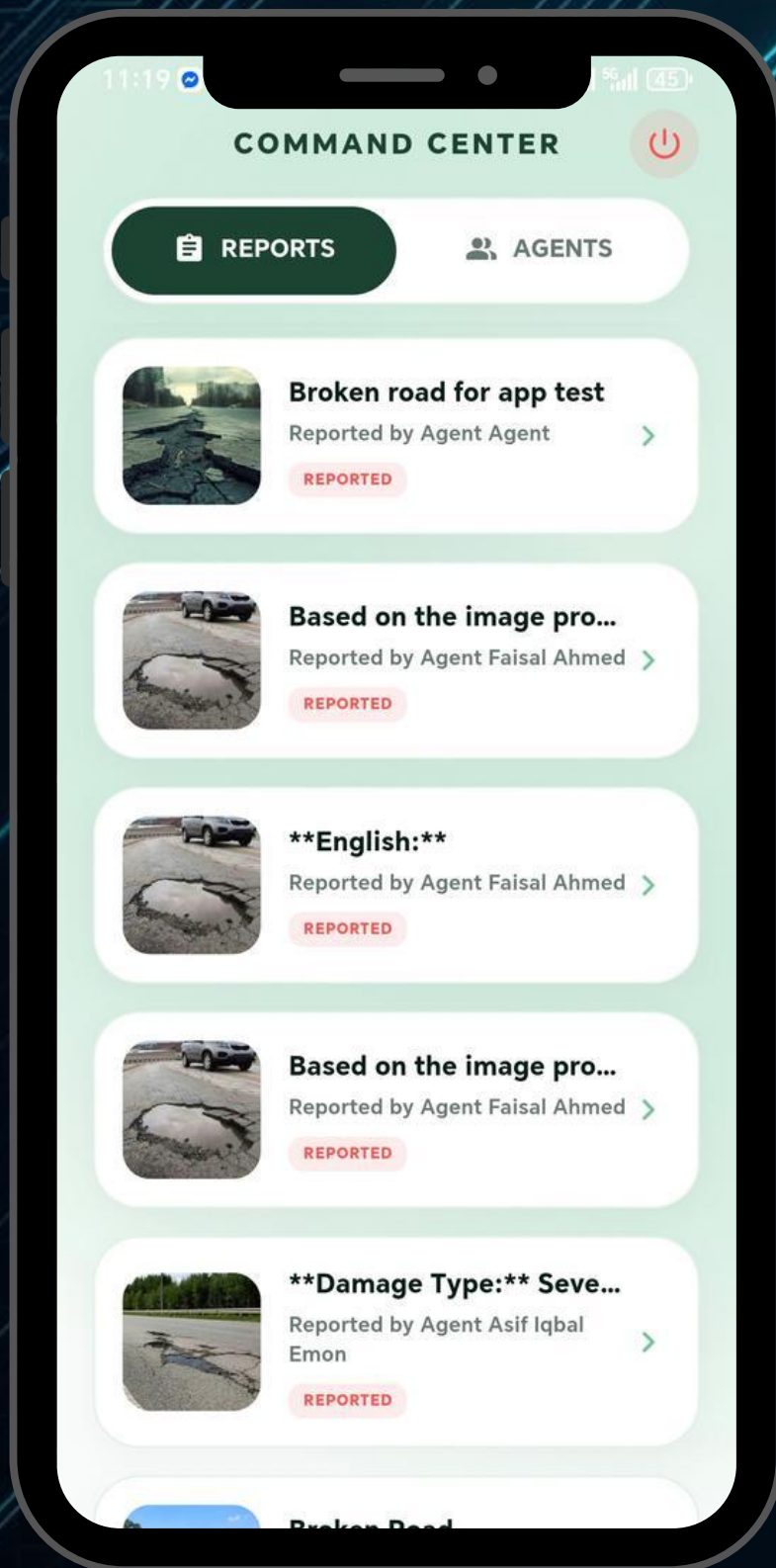
ADMIN PANEL

01

View all hazard reports submitted by community members. Monitor incoming reports in real-time with detailed location data and AI severity analysis.

02

Update hazard status (Reported, In Progress, Resolved). Manage user information and oversee community activity across the platform.





AI INTEGRATION

01

Analyzes road surface conditions using advanced computer vision algorithms to identify potholes, cracks, and damage patterns in real-time

02

Detects and classifies damage severity levels automatically, prioritizing critical hazards that pose immediate safety risks to drivers

03

Improves hazard report accuracy by filtering false positives and validating user submissions with intelligent image analysis

```
1
2  final List<String> API_VERSIONS = ["v1beta", "v1"];
3  final List<String> MODEL_VARIANTS = [
4      "gemini-1.5-flash",
5      "gemini-3-flash-preview",
6      "gemini-2.5-flash",
7      "gemini-1.5-flash-latest",
8      "gemini-pro-vision",
9  ];
10
```




KEY CODE SNIPPETS

01

AI Detection Logic

Core algorithm that analyzes road images and identifies hazard severity using machine learning models.

AI Detection Logic

```
1 Future<AnalysisResult> analyzeRoadHazard({
2     required String base64Image,
3     required Map<String, dynamic> payload,
4 }) async {
5     // Configure Gemini 1.5 Flash Model
6     final model = GenerativeModel(
7         model: 'gemini-1.5-flash',
8         apiKey: payload['apiKey'] as String,
9     );
10
11     // Prepare input parts for multimodal processing
12     final prompt = 'Identify potential hazards and conditions in this road scene.';
13     final imagePart = DataPart('image/png', base64Decode(base64Image));
14     final dataPart = TextPart(jsonEncode(payload['metadata']));
15
16     // Generate content using Gemini AI
17     final response = await model.generateContent([
18         TextPart(prompt),
19         imagePart,
20         dataPart,
21     ]);
22
23     // Process the AI analysis response
24     return AnalysisResult.fromGeminiResponse(response);
25 }
```




KEY CODE SNIPPETS

02

Location Tracking

Real-time GPS integration that auto-captures coordinates when hazards are reported for accurate mapping.

Real-time Location Tracking

```
import 'package:flutter/material.dart';
import 'package:geolocator/geolocator.dart';

Future<Position?> getCurrentLocation() async {
  bool serviceEnabled;
  LocationPermission permission;

  // Check if location services are enabled
  serviceEnabled = await Geolocator.isLocationServiceEnabled();
  if (!serviceEnabled) {
    // Location services are not enabled don't continue
    return null;
  }

  permission = await Geolocator.checkPermission();
  if (permission == LocationPermission.denied) {
    permission = await Geolocator.requestPermission();
    if (permission == LocationPermission.denied) {
      // Permissions are denied, next time you could try
      return null;
    }
  }
  if (permission == LocationPermission.deniedForever) {
    // Permissions are denied forever, handle appropriately.
    return null;
  }
  // When we reach here, permissions are granted and we can
  // continue accessing the position of the device.
  try {
    return await Geolocator.getCurrentPosition(
      desiredAccuracy: LocationAccuracy.high);
  } catch (e) {
    debugPrint(e.toString());
    return null;
  }
}
```




KEY CODE SNIPPETS

03

Verification System

Community voting logic that tracks resolved votes and automatically removes hazards after threshold is met.

Community Verification Logic

```
1 // Function to handle community verification of a road hazard
2 Future<void> verifyHazard(String hazardId) async {
3   final firestore = FirebaseFirestore.instance;
4   final hazardDocRef = firestore.collection('hazards').doc(hazardId);
5
6   try {
7     // Increment 'resolvedVotes' counter atomically using a transaction
8     await firestore.runTransaction((transaction) async {
9       DocumentSnapshot snapshot = await transaction.get(hazardDocRef);
10
11       if (snapshot.exists) {
12         final Map<String, dynamic> data = snapshot.data() as Map<String, dynamic>;
13         int currentResolvedVotes = data['resolvedVotes'] ?? 0;
14         int newResolvedVotes = currentResolvedVotes + 1;
15
16         // Update the resolvedVotes count
17         transaction.update(hazardDocRef, {'resolvedVotes': newResolvedVotes});
18
19         // Check if the hazard has received enough votes to be resolved
20         if (newResolvedVotes >= 3) {
21           // Update status and remove from active map display
22           transaction.update(hazardDocRef, {
23             'status': 'Resolved',
24             'activeOnMap': false,
25           });
26           print('Hazard $hazardId marked as Resolved by community.');
```




KEY FEATURES



AI-Powered Detection

Advanced machine learning algorithms automatically identify and classify road hazards with high accuracy and speed.



Real-Time Map Integration

Live hazard pins on interactive maps help drivers navigate safer routes and avoid dangerous road conditions.



Community Validation

Crowdsourced verification ensures hazard reports are accurate and up-to-date through community voting system.



Admin Monitoring

Comprehensive dashboard for administrators to track reports, manage statuses, and oversee community activity.



FUTURE ROADMAP

01

Government integration for official road maintenance coordination and direct reporting to municipal authorities for faster repairs.

02

Live traffic updates combining hazard data with real-time traffic flow to provide dynamic route optimization and congestion alerts.

03

Advanced AI accuracy improvements and smart notification system for proactive hazard alerts based on user routes and preferences.





CONCLUSION

01

Road Hero significantly improves road safety by empowering communities to identify and report hazards in real-time, creating safer journeys for everyone.

02

By combining advanced AI detection with community-driven verification, Road Hero creates a powerful ecosystem where technology and human insight work together.

03

As a scalable smart city solution, Road Hero is ready for integration with government systems and urban infrastructure to build safer cities worldwide.





Road Hero

Questions?



THANK YOU

